End-Term Examination (Regular & Re-appear)
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: B.Tech CSE-AI/ECE-AI/AI&ML, Semester:1
(December, 2023)

Subject Code: BAI 101	**Subject: Intelligent Systems**
Time :3 Hours	**Maximum Marks :60**

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		**(2.5*8=20)**	
	(a)	Are the statements (P ∨ Q) → R and (P → R) ∨ (Q → R) logically equivalent?	
	(b)	Write the converse, contrapositive and negation of for the following sentence. - If I had $1,000,000, I'd buy you a fur coat	
	(c)	How does the quality and size of a dataset impact the performance of machine learning algorithms?	
	(d)	Can you explain the fundamental concepts behind machine learning and its significance in developing intelligent systems?	
	(e)	What are the key challenges in natural language processing, and how is it applied in chatbots and language translation systems?	
	(f)	How does computer vision contribute to advancements in fields such as healthcare, manufacturing, and security?	
	(g)	Can you highlight the applications of intelligent systems in the development of autonomous vehicles and transportation systems?	
	(h)	How are intelligent systems employed in the field of education to personalize learning experiences and provide adaptive teaching methods?	
		UNIT-i	
Q2	How did the concept of artificial intelligence emerge, and what were some of the early achievements in the field?	**(10)**	
Q3	Describe the significance of knowledge representation in the context of reasoning in intelligent systems.	**(10)**	
		UNIT-II	
Q4	How does pattern recognition play a role in the field of intelligent systems, and what are its applications in real-world scenarios?	**(10)**	
Q5	Why is data considered a crucial asset in the development and implementation of intelligent systems?	**(10)**	
		UNIT-III	
Q6	What are the primary applications of mobile robotics in industries such as logistics, warehousing, and healthcare?	**(10)**	
Q7	How are drones being utilized in various industries, and what role do they play in tasks such as surveillance, agriculture, and disaster response?	**(10)**	
		UNIT-IV	
Q8	How are intelligent systems utilized in the analysis of big data to extract meaningful insights for decision-making in business and research?	**(10)**	
Q9	What applications do intelligent systems have in optimizing agricultural practices and promoting precision farming? Can you provide examples of how intelligent systems are transforming customer service and engagement in various industries?	**(10)**	